

“Yeah, Right...”

Sarcasm Classification on Social Media

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Introduction

Social behavior in humans are a complex subject that can seem almost impossible to fully understand. A good example of this are usage of sarcasm in interactions, which is famously easily to create misunderstandings, especially when written, because of intonation and voice features are often used to clear markers for a sarcastic comment. Combined with a internet where endless of conversations take place everyday about the most obscure topics makes this a complicated and nuanced classification problem. This creates the interesting question: How well can the the machines, which we run these websites on, understand what is really being said? Therefore this project aims to explore how well different machine learning models can classify sarcasm from sentences made on social media.

Methodology

For this project three different models where chosen:

- **Logistic Regression** – To create a good benchmark for the project
- **BiLSTM with multiple head attention** – To explore the connection of differen attention and sarcasm
- **BERTweet**– Showcase to power of fine-tuning complex, pre-trained networks

For the LR and BiLSTM the word embedder FastText was used. This creates a multi-dimensional vector room where every word in the English language gets it’s own representation with a dimension of 300. For the LR-model the word vectors averaged over the whole sentence to get a vector that represents the whole input. This is then fed to the linear layer for it to be transformed by a sigmoid function and then classified by a threshold.

The BiLSTM model takes in a matrix where all the word-vectors are stacked. This is then fed through a LSTM both from the first to last word, and from last to first word, which makes creates two hidden states for every cell. These states are then fed trough a multi-head attention architecture to create a attention matrix, which is then multiplied with the original hidden states to get the attention-based signal. From here signal goes through a final linear layer and then follows the same classification as in the LR.

The BERTweet is a BERT model (Bidirectional Encoder Representations from Transformers) which is first trained on normal language and then fine-tuned on Twitter posts.

Results

Placeholder for results

Conclusion

Data Availability and Report

